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# MASTER'S THESIS ORAL DEFENCE

## Predicting Crowdfunding Campaign Success: A Multimodal Machine Learning & Deep Learning Approach

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# Motivation & Research Gap

## Why predict crowdfunding success?

- Crowdfunding is an increasingly important financing mechanism
- Platforms like Kickstarter connect creators and backers
- Kickstarter hosted:
  - 651k+ projects
  - $8.99B raised
- ~58% of campaigns still fail
- Financial risk for creators and backers
- Early prediction supports decision-making

## Research Gap

- Studies often rely only on metadata or text
- Visual features remain underexplored
- Explainability is often limited
- Many models depend on post-launch dynamics
- Limited comparison across ML, DL and LLM approaches
- LLMs rarely used for structured prediction + reasoning

**This thesis proposes an explainable multimodal framework using only pre-launch information**

# Research Question & Objectives

## How can multimodal data and machine learning techniques be used to effectively predict the success of crowdfunding campaigns using only pre-launch information?

- Compare traditional ML, deep learning and LLM approaches
- Evaluate the predictive value of metadata, text and visual features
- Investigate multimodal fusion strategies
- Compare structured and unstructured text representations
- Investigate explainability and interpretability methods
- Identify the most important predictors of campaign success

# Dataset & Features

## Dataset

- 40,301 Kickstarter campaigns
- Kickstarter platform
- Pre-launch information only
- Binary classification: successful vs unsuccessful

## Structured Metadata

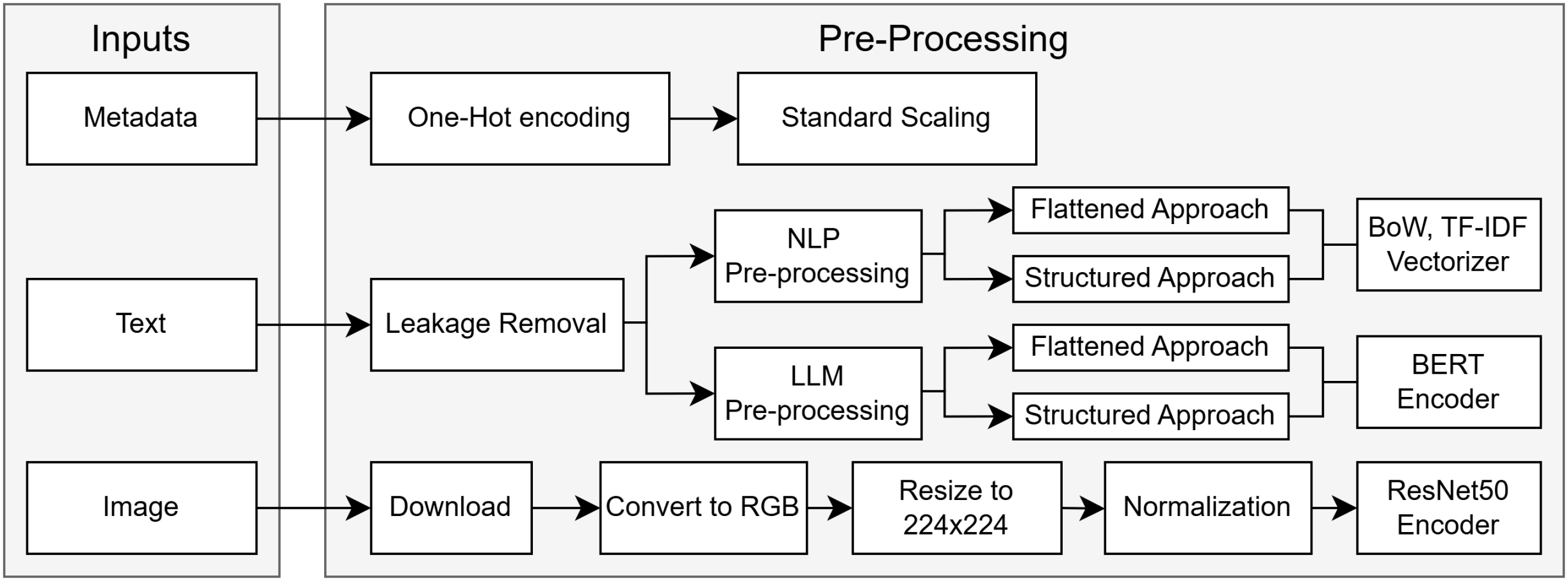
- One-hot encoding
- Feature scaling
- Categorical & numerical campaign attributes

## Textual Features

- Cleaning & tokenization
- Lemmatization & Stopword removal
- TF-IDF & BoW representations
- Sentence-BERT & Fine-tuned BERT embeddings
- Structured and unstructured representations

## Visual Features

- Title image extraction
- Resizing to 224×224
- ResNet50 preprocessing
- ResNet feature extraction
- 2048-dimensional image embeddings


Inputs
Metadata
Text
Image
Pre-Processing
One-Hot encoding
Standard Scaling
Leakage Removal
NLP
Pre-processing
LLM
Pre-processing
Flattened Approach
Structured Approach
Flattened Approach
Structured Approach
BoW, TF-IDF
Vectorizer
BERT
Encoder
Download
Convert to RGB
Resize to
224x224
Normalization
ResNet50
Encoder

# Modeling Framework

## Meta-Only Models

**Inputs**

- Structured metadata only
- Pre-launch campaign attributes

**Models**

- Traditional ML Models
  - Logistic Regression
  - Decision Tree
  - Random Forest
  - SVM
  - XGBoost
  - LightGBM
  - CatBoost
  - KNN

## Metadata + Text Models

**Text Representations**

- TF-IDF
- BoW
- Sentence-BERT
- Fine-tuned BERT

**Text Strategies**

- Structured text
- Unstructured text

**Models**

- Traditional ML models
- Deep learning models
  - Simple Concatenation
  - Deep MLP Fusion
  - Gated Fusion
  - Cross-Interaction
  - Multi-Branch
  - Residual Multi-Branch

## Multimodal Models

**Modalities**

- Metadata
- Text embeddings
- Image embeddings

**Models**

- Deep Learning models
  - Tri-Modal Residual Multi-Branch
  - Tri-Modal Gated Fusion
  - Feature-Level Gated Fusion
- Large Language Model
  - Gemini 2.5 Flash
  - Few-shot prompting
  - Prediction + reasoning

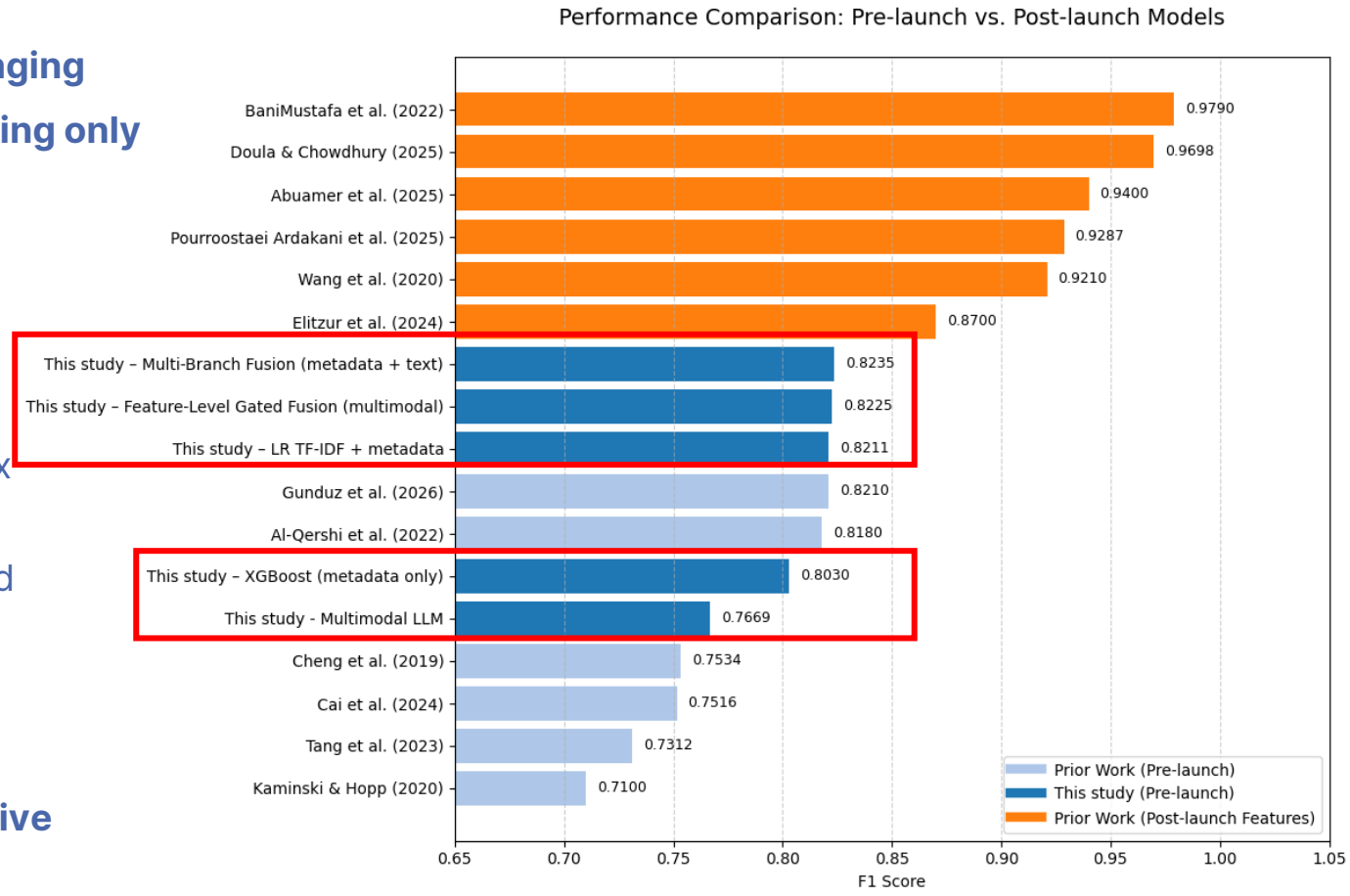
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# Results & Performance Comparison

- **Post-launch models** achieve **high performance** due to near-outcome signals
- **Pre-launch prediction** is significantly more **challenging**
- Proposed models achieved **competitive results using only pre-launch** information
- **Metadata-only models** already achieved **strong performance** (ROC-AUC ~0.88, F1 ~0.79)
- **Textual features** significantly **improved prediction** accuracy (ROC-AUC ~0.90, F1 ~0.82)
- Simpler **linear models outperformed** more complex multimodal architectures
- **Logistic Regression + TF-IDF + metadata** achieved the **best overall performance**

| ROC-AUC | Accuracy | Precision | Recall | F1 |
|---|---|---|---|---|
| 0.904 | 0.816 | 0.82 | 0.82 | 0.82 |

- Visual features provided **limited additional predictive value**
- **LLM did not outperform** traditional ML and DL approaches



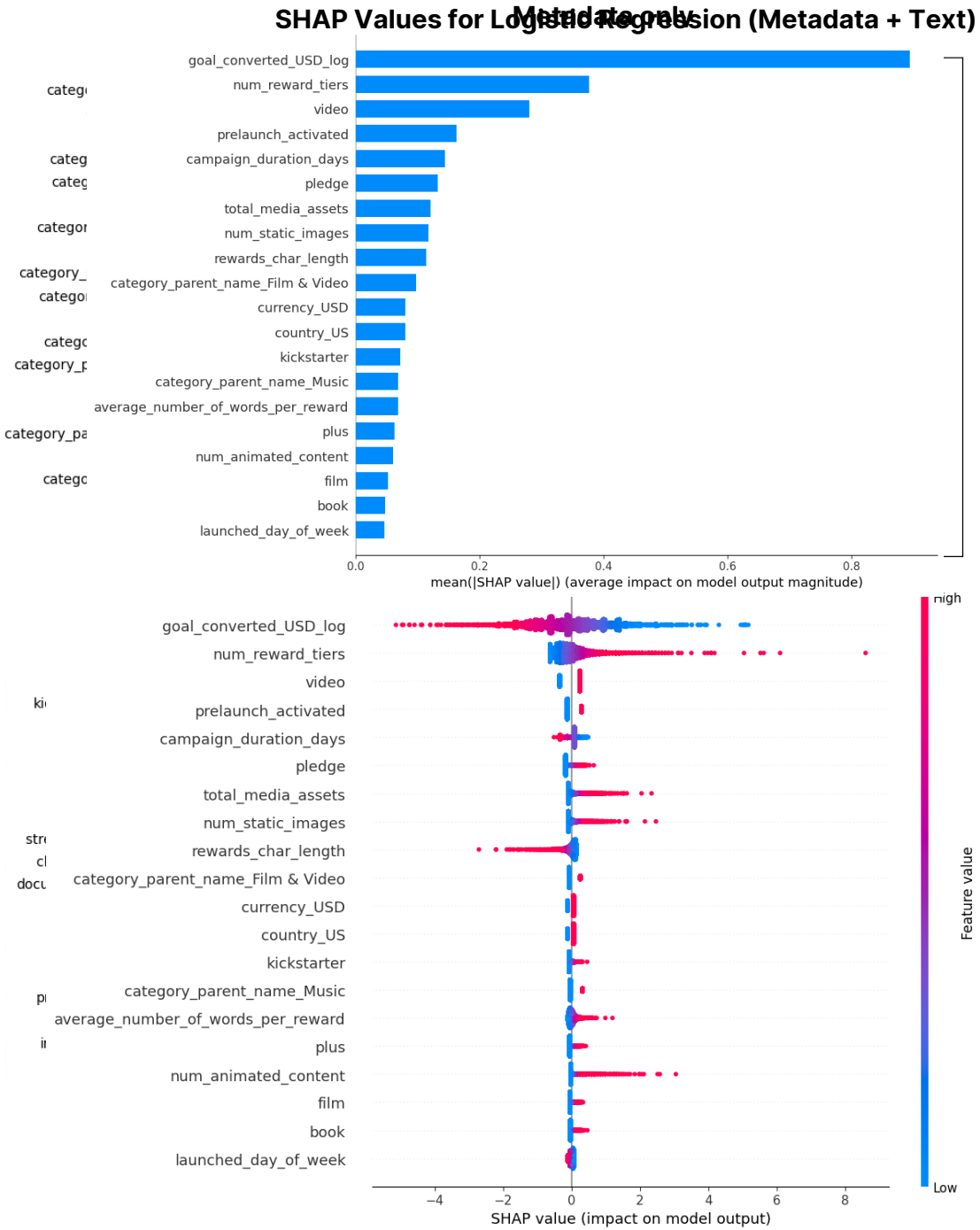
# Explainability & Feature Importance

## Explainability Approaches

- Model coefficients (LR, SVM)
- Tree-based feature importance (XGBoost)
- SHAP values
- LLM-generated qualitative reasoning

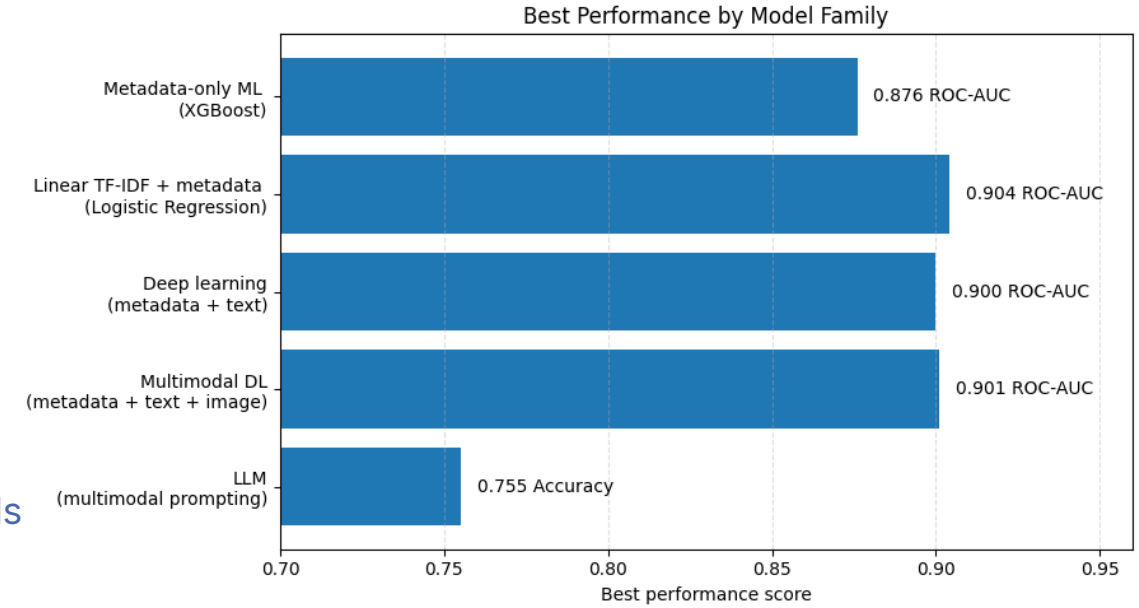
## Insights

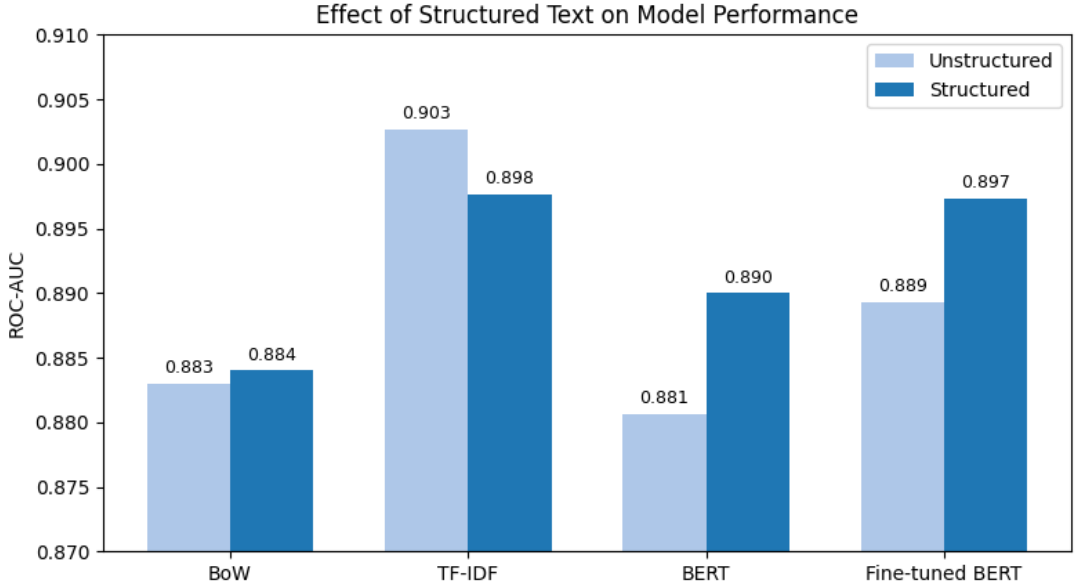
- Raw coefficients were dominated by **one-hot encoded categorical variables** and **high-dimensional text features**
- **SHAP values** provided **more reliable** and direction-aware explanations
- **Metadata features** remained the strongest predictors - **Funding goal, reward tiers, campaign duration, and media assets**
- **LLM explanations provided** valuable **human-readable qualitative reasoning and aligned closely** with SHAP-based findings



# Interpretation & Key Findings

- High-dimensional sparse feature spaces were highly **linearly separable** - linear decision boundaries are particularly effective
- **TF-IDF** representations **outperformed other encoding methods** - **keyword-based features** capture the most predictive signals
- Structured and unstructured representations **captured complementary aspects** of the data - **no consistent dominance** across encoding methods
- **Neural networks handled structured text better** than linear models by learning interactions between text sections
- **DL models** captured richer interactions but provided **only modest performance gains - trade-off**
- Generic pre-trained image embeddings likely limited the value of visual features
- Coefficient-based interpretation became less reliable in high-dimensional encoded feature spaces
- **SHAP values provided more reliable explanations** than coefficient-based interpretation
- LLM-generated reasoning **aligns with quantitative explainability results - can support predictive systems**




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# Conclusion & Contributions

## Main Conclusions

- Crowdfunding success can be predicted effectively using **pre-launch information only**
- Structured metadata alone provided strong predictive performance & Textual features significantly improved predictions
- **Metadata + TF-IDF + Logistic Regression** achieved the **strongest** overall performance
- **More complex deep learning and multimodal models** achieved competitive results, but provided only **limited gains**
- Visual features contributed **limited additional predictive value**
- Incorporating explainability methods **improved transparency and interpretability** of predictions
- LLMs didn't outperform traditional models, but provided **valuable human-readable reasoning**

## Theoretical Contributions

- **Comprehensive comparison** of traditional ML, DL, multimodal and LLM-based architectures
- Comparison of **structured vs unstructured text representations**
- Demonstrated that **feature representation can be more important than model complexity**
- Showed that LLMs can support **prediction + qualitative reasoning**

## Practical Contributions

- Supports **early-stage campaign evaluation**
- **Helps creators optimize** campaign design **before launch**
- Can **support platform recommendation and feedback systems**
- Provides **interpretable predictions** through SHAP and LLM reasoning

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# Limitations

- Visual features were limited to **title images** and **generic pre-trained ResNet embeddings**
- **TF-IDF lacked deeper semantic understanding**, while **BERT fine-tuning was constrained** by dataset size and computational cost
- Structured representations increased sparsity and complexity, especially for linear models
- Analysis **focused exclusively on pre-launch** Kickstarter data, limiting generalizability
- More complex DL and multimodal architectures required **extensive tuning and computational resources**
- Attention-based fusion mechanisms were not explored
- LLM outputs **depended on prompting strategy** and were **not fine-tuned** for the task
- **Explainability methods** were **applied** mainly **to traditional ML** rather than deep learning architectures

# Future Work

- Use **richer visual representations** and multiple campaign images
- Explore **attention-based** and **transformer-based multimodal architectures**
- Incorporate **temporal, social and engagement dynamics**
- Extend analysis **across multiple crowdfunding platforms**
- Further **fine-tune BERT and LLM models** on crowdfunding-specific datasets
- Develop/apply **explainability methods for deep learning** architectures
- Explore greater use of **LLMs with other methods of interpretable AI**

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# THANK YOU FOR YOUR ATTENTION!

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